

CrowdStrike Falcon + Wireshark + Sysmon False Positive Investigation

Suspected C2 beaconing disproved by packet and endpoint evidence

FALSE POSITIVE

Detection defect identified and corrected

Independent evidence review across network, packet, endpoint, and host telemetry.
Complete source evidence is preserved at full size on the following 15 pages.



FALSE POSITIVE

Investigation overview

EVIDENCE TRACK

- CASE BRIEF
- ALERT + RAW EVIDENCE
- PACKET VERIFICATION
- ENDPOINT CORROBORATION
- CORRELATION + JUDGMENT
- DETECTION ENGINEERING
- DISPOSITION + COMMUNICATION

Professional CrowdStrike Falcon + Wireshark + Sysmon False Positive Investigation

Suspected C2 Beaconsing Alert Disproved by Packet and Endpoint Evidence

One high-confidence portfolio scenario showing how a SIEM alert can be correctly investigated and closed as a false positive when the raw network device log, Wireshark capture, CrowdStrike Falcon endpoint telemetry, and Sysmon evidence do not support the alert narrative.

CORE FINDING

Falcon Next-Gen SIEM reported periodic outbound command-and-control beaconsing from FIN-WS-023 to 203.0.113.77. The raw firewall records actually described blocked inbound SYN attempts. A parser/correlation defect reversed directionality and discarded the deny action. Endpoint Wireshark, Falcon EDR, and Sysmon independently showed no communication, execution, persistence, file modification, or registry modification associated with the alleged C2 activity.

Investigation Workflow

1. SIEM Alert	2. Raw Source Validation	3. Wireshark	4. Falcon EDR	5. Sysmon	6. Disposition
High-severity C2 beaconsing	Firewall records are inbound + denied	0 matching frames to IOC	0 endpoint network events to IOC	0 matching network/DNS/change events	FALSE POSITIVE + rule tuning

Professional Use Notes

Note	Professional Handling
Sanitized indicators	All hosts, users, IPs, domains, identifiers, hashes, and logs are synthetic portfolio data. Documentation IP ranges are used for examples.
Negative evidence	Absence in Wireshark is meaningful only because capture coverage, interface, time synchronization, and packet-drop status are documented.
Endpoint corroboration	No single product is treated as absolute proof. The closure is based on agreement among raw firewall logs, packet capture, Falcon endpoint telemetry, and Sysmon.
False-positive definition	The alert narrative asserts outbound C2 from the endpoint, but the underlying event direction is inbound and denied, and no endpoint evidence supports execution or communication.
Detection engineering	The final section corrects the rule by requiring allowed outbound activity and, when attributing behavior to a host process, process-linked endpoint network telemetry.

FALSE POSITIVE

Alert intake and trigger evidence

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1. SIEM Alert Intake and Raw Trigger Evidence

At 12:32:14Z, CrowdStrike Falcon Next-Gen SIEM raises a High alert titled Possible C2 Beaconing - Repetitive HTTPS Sessions. The entity FIN-WS-023 is associated with internal address 10.40.23.55, and the alert claims twelve outbound connections to 203.0.113.77:443 at approximately 60-second intervals.

ANALYST PRINCIPLE

Treat the alert as a hypothesis, not ground truth. The first task is to inspect the raw source events that produced the correlation before assuming the endpoint actually initiated the traffic.

Alert Summary

Field	Value	Analyst Concern
Alert time	2026-08-18T12:32:14.901Z	High-severity correlation fired after 12 repeated events.
SIEM	CrowdStrike Falcon Next-Gen SIEM	Alert is correlation output, not raw endpoint telemetry.
Alert title	Possible C2 Beaconing - Repetitive HTTPS Sessions	Narrative states outbound C2.
Entity	FIN-WS-023 / 10.40.23.55	Host attribution must be independently validated.
Remote IP	203.0.113.77:443	Synthetic external IOC used in the scenario.
Observed count	12 events	Periodicity resembles beaconing.
Mean interval	60.0 seconds	Strong periodicity can be suspicious, but direction and disposition are critical.
Process	Unknown	No process-linked evidence is present in the alert.
Action field	Not carried into correlation	Major red flag: allowed vs denied cannot be determined from alert output.
Direction field	Not carried into correlation	Major red flag: inbound vs outbound cannot be determined from alert output.

Raw Firewall Records That Fed the SIEM

Line	Raw Source	Raw SIEM / Firewall Log	Analyst Read
01	firewall_raw	2026-08-18T12:20:02Z FW-EDGE-01 vendor=ExampleFW direction=inbound src=203.0.113.77 spt=52110 dst=10.40.23.55 dpt=443 proto=tcp flags=SYN action=deny rule=EDGE-IN-DENY session_bytes=0	Blocked inbound SYN; no session established.
02	firewall_raw	2026-08-18T12:21:02Z FW-EDGE-01 vendor=ExampleFW direction=inbound src=203.0.113.77 spt=52147 dst=10.40.23.55 dpt=443 proto=tcp flags=SYN action=deny rule=EDGE-IN-DENY session_bytes=0	Blocked inbound SYN; no session established.
03	firewall_raw	2026-08-18T12:22:02Z FW-EDGE-01 vendor=ExampleFW direction=inbound src=203.0.113.77 spt=52184 dst=10.40.23.55 dpt=443 proto=tcp flags=SYN action=deny rule=EDGE-IN-DENY session_bytes=0	Blocked inbound SYN; no session established.
04	firewall_raw	2026-08-18T12:23:02Z FW-EDGE-01 vendor=ExampleFW direction=inbound src=203.0.113.77 spt=52221 dst=10.40.23.55 dpt=443 proto=tcp flags=SYN action=deny rule=EDGE-IN-DENY session_bytes=0	Blocked inbound SYN; no session established.
05	firewall_raw	2026-08-18T12:24:02Z FW-EDGE-01 vendor=ExampleFW direction=inbound src=203.0.113.77 spt=52258 dst=10.40.23.55 dpt=443 proto=tcp flags=SYN action=deny rule=EDGE-IN-DENY session_bytes=0	Blocked inbound SYN; no session established.

FALSE POSITIVE

Raw records and source validation

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Line	Raw Source	Raw SIEM / Firewall Log	Analyst Read
06	firewall_raw	2026-08-18T12:25:02Z FW-EDGE-01 vendor=ExampleFW direction=inbound src=203.0.113.77 spt=52295 dst=10.40.23.55 dpt=443 proto=tcp flags=SYN action=deny rule=EDGE-IN-DENY session_bytes=0	Blocked inbound SYN; no session established.
07	firewall_raw	2026-08-18T12:26:02Z FW-EDGE-01 vendor=ExampleFW direction=inbound src=203.0.113.77 spt=52332 dst=10.40.23.55 dpt=443 proto=tcp flags=SYN action=deny rule=EDGE-IN-DENY session_bytes=0	Blocked inbound SYN; no session established.
08	firewall_raw	2026-08-18T12:27:02Z FW-EDGE-01 vendor=ExampleFW direction=inbound src=203.0.113.77 spt=52369 dst=10.40.23.55 dpt=443 proto=tcp flags=SYN action=deny rule=EDGE-IN-DENY session_bytes=0	Blocked inbound SYN; no session established.
09	firewall_raw	2026-08-18T12:28:02Z FW-EDGE-01 vendor=ExampleFW direction=inbound src=203.0.113.77 spt=52406 dst=10.40.23.55 dpt=443 proto=tcp flags=SYN action=deny rule=EDGE-IN-DENY session_bytes=0	Blocked inbound SYN; no session established.
10	firewall_raw	2026-08-18T12:29:02Z FW-EDGE-01 vendor=ExampleFW direction=inbound src=203.0.113.77 spt=52443 dst=10.40.23.55 dpt=443 proto=tcp flags=SYN action=deny rule=EDGE-IN-DENY session_bytes=0	Blocked inbound SYN; no session established.
11	firewall_raw	2026-08-18T12:30:02Z FW-EDGE-01 vendor=ExampleFW direction=inbound src=203.0.113.77 spt=52480 dst=10.40.23.55 dpt=443 proto=tcp flags=SYN action=deny rule=EDGE-IN-DENY session_bytes=0	Blocked inbound SYN; no session established.
12	firewall_raw	2026-08-18T12:31:02Z FW-EDGE-01 vendor=ExampleFW direction=inbound src=203.0.113.77 spt=52517 dst=10.40.23.55 dpt=443 proto=tcp flags=SYN action=deny rule=EDGE-IN-DENY session_bytes=0	Blocked inbound SYN; no session established.
13	parser_raw	2026-08-18T12:32:10.202Z Parser=custom_firewall_v4 InputDirection=inbound InputAction=deny OutputLocalIP=10.40.23.55 OutputRemoteIP=203.0.113.77 OutputDirection=<null> OutputAction=<null>	Parser preserved IPs but dropped direction and action.
14	asset_enrichment	2026-08-18T12:32:10.219Z LocalIP=10.40.23.55 Asset=FIN-WS-023 Confidence=high Source=CMDB	Destination host becomes the SIEM entity.
15	correlation_raw	2026-08-18T12:32:14.880Z Rule=Periodic_External_HTTPS GroupBy=Asset,RemoteIP Count=12 AvgInterval=60.0s Threshold=8 DirectionConstraint=false ActionConstraint=false	Rule interprets periodic events without requiring outbound/allowed.
16	alert_raw	2026-08-18T12:32:14.901Z Alert=Possible C2 Beaconing - Repetitive HTTPS Sessions Severity=High Host=FIN-WS-023 LocalIP=10.40.23.55 RemoteIP=203.0.113.77 RemotePort=443 Count=12 Process=unknown	High alert is created from mis-normalized network events.

Packet capture validation

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2. Wireshark Packet Verification - Endpoint Capture

Wireshark is opened on FIN-WS-023 and the full capture covering the alert window is preserved. Because the SIEM claims the endpoint initiated HTTPS C2, the packet capture should contain at minimum an outbound SYN to 203.0.113.77, a completed TCP handshake, or repeated TLS sessions if the alert narrative is correct.

Capture Integrity and Scope

Item	Recorded Value	Why It Matters
Capture host	FIN-WS-023 / 10.40.23.55	Matches the SIEM entity.
Interface	Ethernet 3 - active corporate NIC	Confirms the active path was captured.
Capture window	2026-08-18T12:18:00Z - 12:36:00Z	Starts before the first SIEM event and ends after the alert.
Packets captured	68,412	Sufficient continuous traffic exists to validate visibility.
Capture drops	0 reported	Reduces the chance that all alleged sessions were missed.
NTP drift	< 5 ms against domain time source	Allows direct cross-source timestamp comparison.
Host IP observed	10.40.23.55	Confirms the capture is from the expected endpoint.
Target IOC filter	ip.addr == 203.0.113.77	Returns 0 frames.
Outbound SYN filter	ip.src == 10.40.23.55 && ip.dst == 203.0.113.77 && tcp.flags.syn == 1 && tcp.flags.ack == 0	Returns 0 frames.
DNS IOC filter	dns.qry.name contains "sync-check"	Returns 0 frames.

Raw Wireshark Packet Logs - Normal Endpoint Activity

Line	Frame	Raw Wireshark Packet Log	Packet Interpretation
01	641	2026-08-18T12:18:04.100Z Src=10.40.23.55 Dst=10.40.0.10 Proto=DNS Info=Query A login.microsoftonline.com	Normal DNS request.
02	642	2026-08-18T12:18:04.103Z Src=10.40.0.10 Dst=10.40.23.55 Proto=DNS Info=Response A login.microsoftonline.com=192.0.2.20	Normal DNS response.
03	649	2026-08-18T12:18:04.131Z Src=10.40.23.55 Dst=192.0.2.20 Proto=TCP Info=54318 > 443 SYN	Outbound connection to approved lab SaaS.
04	650	2026-08-18T12:18:04.148Z Src=192.0.2.20 Dst=10.40.23.55 Proto=TCP Info=443 > 54318 SYN,ACK	Handshake response.
05	653	2026-08-18T12:18:04.166Z Src=10.40.23.55 Dst=192.0.2.20 Proto=TLSv1.3 Info=Client Hello SNI=login.microsoftonline.com	Expected encrypted sign-in traffic.
06	912	2026-08-18T12:19:33.018Z Src=10.40.23.55 Dst=10.40.0.10 Proto=DNS Info=Query A officecdn.microsoft.com	Normal Office CDN resolution.
07	913	2026-08-18T12:19:33.021Z Src=10.40.0.10 Dst=10.40.23.55 Proto=DNS Info=Response A officecdn.microsoft.com=192.0.2.31	Normal DNS response.
08	919	2026-08-18T12:19:33.055Z Src=10.40.23.55 Dst=192.0.2.31 Proto=TLSv1.3 Info=Client Hello SNI=officecdn.microsoft.com	Expected Office traffic.
09	1210	2026-08-18T12:20:02.004Z Src=10.40.23.55 Dst=10.40.0.1 Proto=ARP Info=Who has 10.40.0.1? Tell 10.40.23.55	Normal local network activity at first SIEM trigger time.
10	1477	2026-08-18T12:21:01.905Z Src=10.40.23.55 Dst=10.40.1.20 Proto=TCP Info=54881 > 8080 ACK	Connection to corporate proxy, not IOC.
11	1739	2026-08-18T12:22:02.117Z Src=10.40.23.55 Dst=10.40.0.10 Proto=DNS Info=Query A client.wns.windows.com	Normal Windows service DNS.
12	1740	2026-08-18T12:22:02.120Z Src=10.40.0.10 Dst=10.40.23.55 Proto=DNS Info=Response A client.wns.windows.com=192.0.2.44	Normal DNS response.

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Packet evidence and targeted filters

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Wireshark Packet Logs - Continued

Line	Frame	Raw Wireshark Packet Log	Packet Interpretation
13	2044	2026-08-18T12:23:14.884Z Src=10.40.23.55 Dst=10.40.12.25 Proto=SMB2 Info=Session Setup Request account=CORP\mcarter	Normal file-server access.
14	2047	2026-08-18T12:23:14.901Z Src=10.40.12.25 Dst=10.40.23.55 Proto=SMB2 Info=Session Setup Response status=SUCCESS	Normal authenticated SMB.
15	2311	2026-08-18T12:24:02.291Z Src=10.40.23.55 Dst=10.40.0.10 Proto=DNS Info=Query A corp-files.corp.example	Internal DNS lookup.
16	2312	2026-08-18T12:24:02.295Z Src=10.40.0.10 Dst=10.40.23.55 Proto=DNS Info=Response A corp-files.corp.example=10.40.12.25	Internal DNS response.
17	2844	2026-08-18T12:25:55.210Z Src=10.40.23.55 Dst=192.0.2.60 Proto=TLSv1.3 Info=Client Hello SNI=company.sharepoint.com	Expected collaboration traffic.
18	3210	2026-08-18T12:27:02.671Z Src=10.40.23.55 Dst=10.40.0.10 Proto=DNS Info=Query A time.windows.com	Normal time-service resolution.
19	3211	2026-08-18T12:27:02.674Z Src=10.40.0.10 Dst=10.40.23.55 Proto=DNS Info=Response A time.windows.com=192.0.2.70	Normal DNS response.
20	3577	2026-08-18T12:28:30.404Z Src=10.40.23.55 Dst=10.40.1.20 Proto=TCP Info=55214 > 8080 ACK	Normal proxy session continuation.
21	3892	2026-08-18T12:29:02.018Z Src=10.40.23.55 Dst=10.40.0.10 Proto=DNS Info=Query A settings-win.data.microsoft.com	Normal Windows telemetry DNS.
22	3893	2026-08-18T12:29:02.021Z Src=10.40.0.10 Dst=10.40.23.55 Proto=DNS Info=Response A settings-win.data.microsoft.com=192.0.2.81	Normal DNS response.
23	4415	2026-08-18T12:31:02.015Z Src=10.40.23.55 Dst=10.40.0.1 Proto=ARP Info=10.40.23.55 is at 00:11:22:33:44:55	Normal ARP at final SIEM trigger time.
24	5021	2026-08-18T12:32:14.947Z Src=10.40.23.55 Dst=192.0.2.60 Proto=TLSv1.3 Info=Application Data stream=18	Normal existing SharePoint session when alert is created.
25	5884	2026-08-18T12:34:40.111Z Src=10.40.23.55 Dst=10.40.0.10 Proto=DNS Info=Query A ocspp.digicert.com	Normal certificate validation DNS.
26	5885	2026-08-18T12:34:40.114Z Src=10.40.0.10 Dst=10.40.23.55 Proto=DNS Info=Response A ocspp.digicert.com=192.0.2.90	Normal DNS response.

Targeted Wireshark Validation Filters

ip.addr == 203.0.113.77 ip.src == 10.40.23.55 && ip.dst == 203.0.113.77 tcp.flags.syn == 1 && tcp.flags.ack == 0 && ip.src == 10.40.23.55 && tcp.dstport == 443 tls.handshake.type == 1 && ip.dst == 203.0.113.77 dns.qry.name contains "sync-check"		
Filter	Result	Interpretation
ip.addr == 203.0.113.77	0 frames	The IOC never appears on the endpoint capture.
Outbound FIN-WS-023 -> IOC	0 frames	No outbound packet leaves the endpoint for the alleged C2 host.
Outbound SYN to IOC:443	0 frames	No TCP session initiation exists.
TLS Client Hello to IOC	0 frames	No TLS beacon session exists.
DNS query containing sync-check	0 frames	No name-resolution evidence exists for the alleged C2 domain.
PACKET CONCLUSION Validated capture: 0 IOC frames, 0 outbound SYNs, 0 TLS handshakes, and 0 IOC DNS queries. Wireshark directly contradicts the SIEM C2 attribution.		

Falcon endpoint telemetry

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3. CrowdStrike Falcon Endpoint Telemetry - No Corroborating C2

The Falcon sensor is queried independently of the SIEM correlation. CrowdStrike documentation uses ProcessRollup2 for process events and NetworkConnectIP4 for process-linked IPv4 network activity; these are the primary endpoint pivots used here.

Raw Falcon Endpoint Events During the Alert Window

Line	Falcon Event	Raw Falcon Event / Query Result	Analyst Read
01	ProcessRollup2	2026-08-18T12:18:01.442Z ComputerName=FIN-WS-023 aid=fp001 ImageFileName=C:\Windows\explorer.exe ParentBaseFileName=userinit.exe CommandLine=explorer.exe TargetProcessId=4420	Expected shell process.
02	ProcessRollup2	2026-08-18T12:18:02.119Z ComputerName=FIN-WS-023 aid=fp001 ImageFileName=C:\Program Files\Microsoft Office\root\Office16\OUTLOOK.EXE ParentBaseFileName=explorer.exe CommandLine=OUTLOOK.EXE TargetProcessId=6152	Expected Outlook process.
03	ProcessRollup2	2026-08-18T12:18:03.008Z ComputerName=FIN-WS-023 aid=fp001 ImageFileName=C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe ParentBaseFileName=explorer.exe CommandLine=msedge.exe TargetProcessId=7004	Expected browser process.
04	NetworkConnectIP4	2026-08-18T12:18:04.131Z ComputerName=FIN-WS-023 aid=fp001 ContextBaseFileName=msedge.exe ContextProcessId=7004 LocalAddressIP4=10.40.23.55 RemoteAddressIP4=192.0.2.20 RemotePort=443	Matches normal Wireshark connection.
05	NetworkConnectIP4	2026-08-18T12:19:33.055Z ComputerName=FIN-WS-023 aid=fp001 ContextBaseFileName=OUTLOOK.EXE ContextProcessId=6152 LocalAddressIP4=10.40.23.55 RemoteAddressIP4=192.0.2.31 RemotePort=443	Expected Office traffic.
06	UserLogon	2026-08-18T12:20:11.228Z ComputerName=FIN-WS-023 aid=fp001 UserName=mcarter LogonType=Interactive	Normal user session.
07	ProcessRollup2	2026-08-18T12:22:41.910Z ComputerName=FIN-WS-023 aid=fp001 ImageFileName=C:\Program Files\Microsoft OneDrive\OneDrive.exe ParentBaseFileName=explorer.exe CommandLine=OneDrive.exe /background TargetProcessId=8040	Expected sync client.
08	NetworkConnectIP4	2026-08-18T12:25:55.210Z ComputerName=FIN-WS-023 aid=fp001 ContextBaseFileName=OneDrive.exe ContextProcessId=8040 LocalAddressIP4=10.40.23.55 RemoteAddressIP4=192.0.2.60 RemotePort=443	Matches normal collaboration traffic.
09	ProcessRollup2	2026-08-18T12:27:58.090Z ComputerName=FIN-WS-023 aid=fp001 ImageFileName=C:\Windows\System32\svchost.exe ParentBaseFileName=services.exe CommandLine=svchost.exe -k netsvcs TargetProcessId=1160	Expected Windows service host.
10	NetworkConnectIP4	2026-08-18T12:29:02.044Z ComputerName=FIN-WS-023 aid=fp001 ContextBaseFileName=svchost.exe ContextProcessId=1160 LocalAddressIP4=10.40.23.55 RemoteAddressIP4=192.0.2.81 RemotePort=443	Normal Windows service traffic.
11	ProcessRollup2	2026-08-18T12:31:48.337Z ComputerName=FIN-WS-023 aid=fp001 ImageFileName=C:\Windows\System32\SearchHost.exe ParentBaseFileName=svchost.exe CommandLine=SearchHost.exe TargetProcessId=9228	Expected Windows process.
12	NetworkConnectIP4	2026-08-18T12:32:14.947Z ComputerName=FIN-WS-023 aid=fp001 ContextBaseFileName=OneDrive.exe ContextProcessId=8040 LocalAddressIP4=10.40.23.55 RemoteAddressIP4=192.0.2.60 RemotePort=443	Normal session at alert time.
13	hunt_result	2026-08-18T12:35:10.100Z Query=NetworkConnectIP4 ComputerName=FIN-WS-023 RemoteAddressIP4=203.0.113.77 ResultCount=0	No Falcon network event to IOC.
14	hunt_result	2026-08-18T12:35:22.410Z Query=ProcessRollup2 joined NetworkConnectIP4 RemoteAddressIP4=203.0.113.77 ResultCount=0	No process can be tied to the alleged C2.

FALSE POSITIVE

Falcon pivots and result matrix

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Line	Falcon Event	Raw Falcon Event / Query Result	Analyst Read
15	sensor_health	2026-08-18T12:35:31.008Z ComputerName=FIN-WS-023 aid=fp001 SensorState=online EventFlow=normal	Sensor was online during the window.

Falcon CQL Validation Pivots

```
#event_simpleName=NetworkConnectIP4 ComputerName="FIN-WS-023" RemoteAddressIP4="203.0.113.77"
| table([@timestamp, ComputerName, ContextBaseFileName, LocalAddressIP4, RemoteAddressIP4, RemotePort])

#event_simpleName=ProcessRollup2 ComputerName="FIN-WS-023"
| join({#event_simpleName=NetworkConnectIP4 RemoteAddressIP4="203.0.113.77"},
      key=ContextProcessId, field=TargetProcessId,
      include=[RemoteAddressIP4, RemotePort])
| table([@timestamp, ImageFileName, CommandLine, ParentBaseFileName, RemoteAddressIP4, RemotePort])
```

Falcon Verification Result Matrix

Validation	Result	Meaning
NetworkConnectIP4 to 203.0.113.77	0 events	Falcon did not observe an endpoint connection to the IOC.
ProcessRollup2 -> NetworkConnectIP4 join	0 correlated rows	No process can be linked to the alleged C2 channel.
Normal network events during window	Present	The sensor was collecting network telemetry normally.
Normal process events during window	Present	Process visibility was functioning during the alert.
Sensor health	Online / event flow normal	Absence is not explained by a disconnected sensor.
Cross-check with Wireshark	Matches	Falcon and packet evidence agree on the endpoint network state.

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4. Sysmon Validation - No Network, DNS, File, or Registry Evidence

Sysmon is used as a second endpoint evidence source. Microsoft documents Event ID 1 for process creation and Event ID 3 for network connections; the configured lab policy also records DNS Query events (Event ID 22), FileCreate events (Event ID 11), and registry value changes such as Event ID 13.

Raw Sysmon Events During the Alert Window

Line	Sysmon Source	Raw Sysmon Log / Query Result	Analyst Read
01	1 - Process Create	2026-08-18 12:18:02.116Z Image=C:\Program Files\Microsoft Office\root\Office16\OUTLOOK.EXE ParentImage=C:\Windows\explorer.exe User=CORP\mcarter ProcessGuid={FP-6152}	Expected process.
02	1 - Process Create	2026-08-18 12:18:03.006Z Image=C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe ParentImage=C:\Windows\explorer.exe User=CORP\mcarter ProcessGuid={FP-7004}	Expected browser.
03	22 - DNS Query	2026-08-18 12:18:04.098Z Image=msedge.exe ProcessGuid={FP-7004} QueryName=login.microsoftonline.com QueryStatus=0	Matches Wireshark DNS.
04	3 - Network Connect	2026-08-18 12:18:04.130Z Image=msedge.exe ProcessGuid={FP-7004} SourceIp=10.40.23.55 SourcePort=54318 DestinationIp=192.0.2.20 DestinationPort=443 Protocol=tcp Initiated=true	Matches Wireshark + Falcon.
05	22 - DNS Query	2026-08-18 12:19:33.016Z Image=OUTLOOK.EXE ProcessGuid={FP-6152} QueryName=officecdn.microsoft.com QueryStatus=0	Normal Office DNS.
06	3 - Network Connect	2026-08-18 12:19:33.054Z Image=OUTLOOK.EXE ProcessGuid={FP-6152} SourceIp=10.40.23.55 DestinationIp=192.0.2.31 DestinationPort=443 Protocol=tcp Initiated=true	Normal Office network activity.
07	1 - Process Create	2026-08-18 12:22:41.908Z Image=C:\Program Files\Microsoft OneDrive\OneDrive.exe ParentImage=C:\Windows\explorer.exe User=CORP\mcarter ProcessGuid={FP-8040}	Expected OneDrive process.
08	22 - DNS Query	2026-08-18 12:25:55.202Z Image=OneDrive.exe ProcessGuid={FP-8040} QueryName=company.sharepoint.com QueryStatus=0	Normal SharePoint DNS.
09	3 - Network Connect	2026-08-18 12:25:55.209Z Image=OneDrive.exe ProcessGuid={FP-8040} SourceIp=10.40.23.55 DestinationIp=192.0.2.60 DestinationPort=443 Protocol=tcp Initiated=true	Matches normal Wireshark + Falcon traffic.
10	1 - Process Create	2026-08-18 12:27:58.087Z Image=C:\Windows\System32\svchost.exe ParentImage=C:\Windows\System32\services.exe User=NT AUTHORITY\SYSTEM ProcessGuid={FP-1160}	Expected Windows service process.
11	22 - DNS Query	2026-08-18 12:29:02.016Z Image=svchost.exe ProcessGuid={FP-1160} QueryName=settings-win.data.microsoft.com QueryStatus=0	Normal Windows DNS.
12	3 - Network Connect	2026-08-18 12:29:02.042Z Image=svchost.exe ProcessGuid={FP-1160} SourceIp=10.40.23.55 DestinationIp=192.0.2.81 DestinationPort=443 Protocol=tcp Initiated=true	Normal service network activity.
13	query_result	2026-08-18 12:35:44.100Z EventID=3 DestinationIp=203.0.113.77 ResultCount=0	No Sysmon network connection to IOC.
14	query_result	2026-08-18 12:35:48.422Z EventID=22 QueryName contains sync-check ResultCount=0	No DNS query associated with alleged C2.
15	query_result	2026-08-18 12:35:55.011Z EventID=11 SuspiciousPathScope=Temp,AppData,ProgramData ResultCount=0	No suspicious file creation tied to alert.
16	query_result	2026-08-18 12:35:59.334Z EventID=13 PersistenceKeyScope=Run,RunOnce,Services ResultCount=0	No persistence-oriented registry modification tied to alert.

ENDPOINT CONCLUSION

Falcon and Sysmon agree with Wireshark. The host shows normal processes and normal network activity, but there is no network event to 203.0.113.77, no DNS query for the alleged C2 infrastructure, no suspicious process chain, and no file or registry change associated with the alert.

FALSE POSITIVE

Cross-source timeline

EVIDENCE TRACK

- CASE BRIEF
- ALERT + RAW EVIDENCE
- PACKET VERIFICATION
- ENDPOINT CORROBORATION
- CORRELATION + JUDGMENT
- DETECTION ENGINEERING
- DISPOSITION + COMMUNICATION

5. Normalized Cross-Source Investigation Timeline

The normalized timeline puts the alert, raw firewall events, packet evidence, Falcon endpoint telemetry, and Sysmon evidence on one clock. The key investigative pattern is not merely “nothing found”; it is a direct contradiction between the SIEM narrative and the underlying source direction/disposition.

Line	Time (Z)	Evidence Source	Key Evidence	What It Proves
01	12:20:02	Firewall raw	203.0.113.77 -> 10.40.23.55:443 SYN action=deny direction=inbound	Blocked inbound attempt; endpoint did not initiate it.
02	12:20:02	Wireshark	Normal ARP only; no 203.0.113.77 frame	No corresponding packet reaches/leaves endpoint.
03	12:21:02	Firewall raw	Second inbound denied SYN	Periodic pattern begins.
04	12:21:02	Wireshark	Corporate proxy ACK; no IOC traffic	Endpoint continues normal activity.
05	12:22:02	Firewall raw	Third inbound denied SYN	Still denied at firewall.
06	12:22:02	Wireshark/Sysmon	Normal Windows DNS; no IOC	Independent endpoint sources do not corroborate.
07	12:25:55	Falcon/Sysmon/Wireshark	OneDrive -> approved lab SaaS 192.0.2.60:443	Three sources align on legitimate connection.
08	12:29:02	Falcon/Sysmon/Wireshark	svchost -> 192.0.2.81:443	Three sources again align on legitimate traffic.
09	12:31:02	Firewall raw	Final inbound denied SYN in series	No successful session established.
10	12:31:02	Wireshark	ARP only; no IOC	Final alleged beacon has no endpoint packet.
11	12:32:10	Parser	Direction/action discarded during normalization	Critical data-quality failure introduced.
12	12:32:10	Asset enrichment	10.40.23.55 mapped to FIN-WS-023	Target of inbound traffic becomes alert entity.
13	12:32:14	SIEM correlation	12 events / avg interval 60s / no direction or action constraint	Periodicity alone becomes C2 hypothesis.
14	12:32:14	SIEM alert	High: Possible C2 Beaconsing	Alert narrative overstates what the source records prove.
15	12:34:50	Wireshark pivot	ip.addr == 203.0.113.77 -> 0 frames	Packet evidence contradicts alert attribution.
16	12:35:10	Falcon CQL	NetworkConnectIP4 to IOC -> 0 events	EDR contradicts endpoint connection claim.
17	12:35:44	Sysmon	Event ID 3 to IOC -> 0 events	Second endpoint source contradicts connection claim.
18	12:35:48	Sysmon	Event ID 22 IOC DNS -> 0 events	No name-resolution path exists.
19	12:36:20	Analyst	Raw firewall direction/action validated	Original traffic is inbound and denied.
20	12:38:00	Disposition	FALSE POSITIVE - detection logic / parser defect	Incident closed; detection engineering ticket opened.

Evidence Conflict Matrix

Question	SIEM Alert Claims	Independent Evidence	Judgment
Who initiated the traffic?	FIN-WS-023	Raw firewall: 203.0.113.77 initiated inbound SYNs	SIEM direction is wrong.
Was the traffic allowed?	Not stated	Raw firewall: action=deny, session_bytes=0	No session established.
Did packets reach/leave endpoint?	Implied yes	Wireshark: 0 IOC frames	No packet corroboration.
Did Falcon see a connection?	Implied endpoint C2	NetworkConnectIP4 to IOC: 0	No EDR corroboration.
Did Sysmon see a connection?	Implied endpoint C2	Event ID 3 to IOC: 0	No host network corroboration.
Was there DNS resolution?	Unknown	Wireshark + Sysmon DNS: 0 IOC queries	No domain-resolution evidence.
Did process/file/registry state change?	Unknown	No suspicious process chain; no EID 11/13 match	No endpoint effect associated with alert.

FALSE POSITIVE

Analyst judgment and evidence priority

EVIDENCE TRACK

- CASE BRIEF
- ALERT + RAW EVIDENCE
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6. Clean Line-by-Line Analyst Interpretation and SOC Judgment

Line	Analyst Identification	Why It Matters / Next Pivot	Priority
01	SIEM detects 60-second periodicity	Periodicity is a valid hunting signal but does not establish direction or successful communication.	High
02	Raw firewall logs show inbound SYN	The external host is the initiator. This directly conflicts with the alert narrative.	Critical
03	Firewall action is deny	The security control blocks the attempt before a session is established.	Critical
04	Firewall session_bytes=0	No payload exchange is recorded by the source device.	High
05	Parser drops action	The normalized event loses the field required to distinguish allowed traffic from blocked probes.	Critical
06	Parser drops direction	The normalized event loses the field required to distinguish inbound scanning from outbound C2.	Critical
07	Asset enrichment maps destination IP to host	The host becomes the alert entity even though it is the target, not the source.	High
08	Wireshark capture is continuous with zero drops	Negative packet evidence is trustworthy enough to use in correlation.	High
09	Wireshark returns 0 IOC frames	No inbound or outbound IOC traffic is visible at the endpoint.	Critical
10	Falcon NetworkConnectIP4 returns 0	No Falcon-sensor network event supports the claimed connection.	Critical
11	Falcon process-network join returns 0	No process can be associated with the alleged C2 channel.	Critical
12	Sysmon Event ID 3 returns 0	A second endpoint telemetry source also sees no connection.	Critical
13	Sysmon Event ID 22 returns 0	No DNS resolution supports the alleged destination/domain.	High
14	Sysmon EID 11/13 targeted searches return 0	No suspicious file or persistence registry change accompanies the alert.	High
15	Normal Falcon/Sysmon/Wireshark events align	Cross-source visibility is working; the lack of IOC evidence is not caused by general telemetry failure.	Critical
16	Alert lacks process identity	The rule cannot establish which executable supposedly generated the beacon.	High
17	Raw source contradicts normalized narrative	This is a data-normalization/detection defect, not ambiguous endpoint behavior.	Critical
18	SOC disposition	Close as false positive and open detection-engineering remediation rather than host containment.	Critical

Evidence Priority

Priority	Evidence	Reason
1	Raw firewall lines 01-12	They are the original source records and prove direction=inbound, action=deny, session_bytes=0.
2	Wireshark capture + integrity metadata	The exact endpoint/window contains normal traffic but no IOC frames.
3	Falcon NetworkConnectIP4 + process join	Endpoint sensor shows no connection and no process-linked network path to the IOC.
4	Sysmon Event IDs 3 and 22	Independent OS telemetry shows neither connection nor DNS resolution.
5	Sysmon Event IDs 11 and 13	No file or persistence-registry effect supports compromise.
6	Parser/correlation output	Explains how the raw truth was transformed into an incorrect C2 narrative.

- CASE BRIEF
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7. Investigation Pivots and Detection Engineering Correction

ANALYST CONCLUSION

FALSE POSITIVE. Raw firewall evidence proves inbound denied traffic; Wireshark, Falcon EDR, and Sysmon show no endpoint connection or malicious state change. The root cause is a parser/correlation defect that dropped direction and action before periodicity analysis.

CrowdStrike Falcon CQL - Confirm No Endpoint Connection to IOC

```
#event_simpleName=NetworkConnectIP4 ComputerName="FIN-WS-023" RemoteAddressIP4="203.0.113.77"  
| table([@timestamp, ComputerName, ContextBaseFileName, ContextProcessId, LocalAddressIP4, RemoteAddressIP4, RemotePort])
```

CrowdStrike Falcon CQL - Require Process/Network Correlation

```
#event_simpleName=ProcessRollup2 ComputerName="FIN-WS-023"  
| join({#event_simpleName=NetworkConnectIP4 RemoteAddressIP4="203.0.113.77"},  
      key=ContextProcessId, field=TargetProcessId,  
      include=[RemoteAddressIP4, RemotePort])  
| table([@timestamp, ImageFileName, CommandLine, ParentBaseFileName, RemoteAddressIP4, RemotePort])
```

Wireshark

```
ip.addr == 203.0.113.77  
ip.src == 10.40.23.55 && ip.dst == 203.0.113.77  
tcp.flags.syn == 1 && tcp.flags.ack == 0 && ip.src == 10.40.23.55 && tcp.dstport == 443  
tls.handshake.type == 1 && ip.dst == 203.0.113.77
```

Sysmon Targeted Review

```
# Review the Sysmon Operational log over the exact alert window.  
# Required pivots: Event ID 3 DestinationIp=203.0.113.77; Event ID 22 IOC domain;  
# Event ID 1 suspicious process lineage; Event ID 11 file creation; Event ID 13 registry value changes.
```

Corrected Correlation Logic - Design Requirements

Control	Old Behavior	Corrected Requirement
Direction	Ignored	Require normalized direction=outbound for a C2 beacon rule.
Action / disposition	Ignored	Require allow/accept/success; exclude deny/drop/reset-only events.
Session establishment	Not checked	Prefer established-session evidence or bytes transferred, not isolated SYN attempts.
Host attribution	Mapped any internal IP to asset	Use source/local initiator semantics; do not treat an inbound destination as the originator.
Process corroboration	Not required	When claiming endpoint C2, correlate to process-linked Falcon network telemetry when available.
Raw field preservation	Direction/action discarded	Preserve raw + normalized direction and disposition for analyst review.
Periodicity	Primary decision factor	Use periodicity as a supporting signal after direction/session validity is established.
Inbound scans	Mixed into C2 logic	Route repeated denied inbound SYNs to a separate reconnaissance/scan analytic.

FALSE POSITIVE

Detection logic and regression testing

EVIDENCE TRACK

- CASE BRIEF
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Conceptual Corrected Rule

```
// Field names vary by firewall parser. This is detection logic, not a vendor-specific parser contract.  
network.direction = "outbound"  
AND event.action IN ("allow","accept","success")  
AND destination.port IN (80,443)  
AND destination.ip is external  
AND session.count >= 8 within 15m  
AND periodicity_score >= threshold  
AND (process-linked endpoint connection exists OR network session has established/byte evidence)  
// Explicitly exclude deny/drop-only inbound probes.
```

Regression Test Suite for the Corrected Analytic

Test Case	Input	Expected Result
A - This false positive	12 inbound denied SYNs at 60-second intervals	C2 rule does NOT fire; optional inbound-scan rule may fire.
B - Real outbound beacon	12 allowed outbound sessions at 60-second intervals with established bytes	C2 rule fires.
C - Blocked outbound attempts	12 outbound denies with zero session bytes	C2 rule suppressed or downgraded; investigate policy separately.
D - Allowed browser traffic	Periodic HTTPS from signed browser to approved service	C2 rule suppressed by reputation/baseline/process context.
E - Endpoint corroboration missing	Network source claims outbound C2 but Falcon process/network evidence is absent	Alert marked low-confidence / needs validation rather than auto-escalated.
F - Parser field loss	Direction or action becomes null during normalization	Data-quality alert/tuning test fails closed instead of producing C2.

FALSE POSITIVE

Disposition and response plan

EVIDENCE TRACK

- CASE BRIEF
- ALERT + RAW EVIDENCE
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8. False Positive Closure, Remediation, and Portfolio Explanation

SOC Disposition Record

Field	Final Value
Classification	FALSE POSITIVE
Alert	Possible C2 Beaconsing - Repetitive HTTPS Sessions
Affected entity	FIN-WS-023 / 10.40.23.55
Alleged IOC	203.0.113.77:443
Actual network behavior	Repeated inbound TCP SYN attempts blocked by FW-EDGE-01; action=deny; session_bytes=0.
Wireshark	0 matching frames to/from IOC during complete validated capture window.
CrowdStrike Falcon	0 NetworkConnectIP4 events to IOC; 0 process-network joins to IOC; sensor online.
Sysmon	0 Event ID 3 to IOC; 0 IOC DNS queries; no suspicious EID 11/13 state change tied to alert.
Root cause	Custom firewall normalization dropped direction/action; correlation rule did not require outbound allowed sessions.
Host containment	Not required based on evidence.
Detection engineering	Required - parser fix, rule constraints, regression test, separate inbound scan analytic.
Confidence	High - multiple independent sources agree and raw source contradicts the alert narrative.

Remediation Plan

Phase	Action	Expected Result
1. Preserve evidence	Save original SIEM alert, raw firewall events, parser output, PCAP metadata/hash, Falcon query results, and Sysmon export.	Creates reproducible false-positive evidence.
2. Fix parser	Restore direction and action/disposition fields from the firewall source.	Prevents blocked inbound traffic from being normalized as generic host communication.
3. Fix analytic	Require outbound + allowed/established activity before calculating beacon periodicity.	Removes the logical condition that created the false alert.
4. Add endpoint corroboration	For endpoint C2 claims, enrich with Falcon process-linked NetworkConnectIP4 evidence where available.	Improves entity and process confidence.
5. Regression test	Replay the twelve denied inbound SYN records and confirm the C2 rule does not fire.	Proves the defect is fixed.
6. Preserve scan visibility	Create/tune a separate inbound reconnaissance rule for repeated denied probes.	Retains useful detection without misclassifying it as endpoint C2.
7. Monitor	Watch for recurrence of direction/action-null events and false host attribution.	Validates parser and rule health over time.

Do Not Close as False Positive Until

Criterion	Required Evidence
Raw source validated	Original firewall record confirms inbound direction and denied/drop disposition.
Packet visibility validated	Correct endpoint/interface/window captured with trustworthy time alignment and no material drops.
Endpoint network checked	Falcon and Sysmon show no matching connection to the IOC.
Execution checked	No suspicious process lineage exists that could explain the alert.
Effect checked	No related file or registry modification indicates persistence or payload activity.
Telemetry health checked	Normal events are present in Falcon, Sysmon, and Wireshark, proving visibility was functioning.
Root cause identified	Parser/correlation defect is documented rather than simply stating “no evidence found.”

FALSE POSITIVE

Professional case explanation

EVIDENCE TRACK

- CASE BRIEF
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Clean Professional Explanation

The SIEM raised a high-severity C2 beaconing alert because it observed twelve events involving FIN-WS-023 and 203.0.113.77 at almost exactly 60-second intervals. Periodicity made the event pattern look like malware beaconing, but the alert did not preserve network direction or firewall disposition and did not identify a responsible endpoint process.

Review of the original firewall records changed the interpretation immediately. Every event was an inbound SYN from 203.0.113.77 to 10.40.23.55, and every attempt was denied with zero session bytes. The custom parser preserved the two IP addresses but discarded the inbound direction and deny action. Asset enrichment then mapped the internal destination to FIN-WS-023, and the correlation rule incorrectly treated the target system as the source of periodic outbound activity.

Wireshark captured the active endpoint interface across the entire alert window with normal traffic and no reported drops. Targeted filters returned zero frames to or from 203.0.113.77, including zero outbound SYNs and zero TLS Client Hello packets. CrowdStrike Falcon independently returned zero NetworkConnectIP4 events to the IOC and no ProcessRollup2-to-network correlation. Sysmon also returned zero Event ID 3 network connections to the IOC and no matching DNS query, suspicious file creation, or persistence registry modification.

The incident is therefore closed as a false positive with high confidence. The key lesson is that a SIEM alert is an investigative hypothesis. A confident false-positive decision comes from explaining exactly why the detection fired, proving that the raw source does not support the alert narrative, and demonstrating that independent packet and endpoint telemetry agree with the corrected interpretation.

EVIDENCE TRACK

- CASE BRIEF
- ALERT + RAW EVIDENCE
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9. Portfolio Discussion Points and Technical Accuracy Notes

Portfolio Discussion Points

Question	Professional Answer Focus
Why was this a false positive?	The SIEM narrative said outbound C2, but raw firewall events were inbound and denied, and three independent endpoint evidence sources did not corroborate the connection.
Why use Wireshark?	It validates whether the endpoint actually transmitted or received the alleged traffic and provides timing, direction, handshakes, streams, and packet-level evidence.
Why is “0 packets” not automatically proof?	Negative packet evidence is only useful after validating capture interface, time window, drops, host IP, and sensor placement.
What did Falcon add?	Falcon provided process-linked endpoint network telemetry; no NetworkConnectIP4 event or process correlation existed for the IOC.
What did Sysmon add?	Sysmon independently showed no network connection, no IOC DNS query, and no related file/registry effect.
What actually caused the alert?	A normalization defect dropped direction/action fields, and a periodicity rule treated an inbound destination as an outbound source.
What was the remediation?	Fix the parser, require outbound allowed/established sessions, add process corroboration, regression-test the rule, and separate inbound scan detections from C2 detections.
What does this demonstrate?	SOC triage, packet analysis, EDR hunting, Sysmon validation, cross-source correlation, false-positive closure discipline, and detection engineering.

Technical Accuracy Notes

CrowdStrike Falcon Next-Gen SIEM supports CQL searching and official CrowdStrike examples correlate ProcessRollup2 process events with NetworkConnectIP4 network events using process identifiers. The exact field set available in a real tenant depends on product licensing, sensor/platform, repository, and ingestion configuration. Microsoft Sysmon documents Event ID 1 as Process Create and Event ID 3 as Network Connection. Sysmon event coverage is configuration-dependent; network connection logging can be disabled, so an analyst must confirm the deployed Sysmon configuration before using absence as evidence. The firewall vendor and parser in this scenario are intentionally generic. Real third-party field names vary. The important detection-engineering principle is to preserve source direction and disposition and to avoid converting blocked inbound probes into outbound endpoint communications.

Reference Notes

Reference	Use in This Scenario
CrowdStrike Developer Documentation - ngsiem_search	Validates CQL search capability and official examples joining ProcessRollup2 with NetworkConnectIP4 for process/network investigation.
CrowdStrike technical threat-hunting publications	Show current Falcon LogScale / Next-Gen SIEM use of #event_simpleName=ProcessRollup2 and NetworkConnectIP4 in detection queries.
Microsoft Sysinternals - Sysmon	Validates Sysmon process and network event semantics and the need to account for configuration.
Supplied AD FS Golden SAML portfolio PDF	Used as the structural model for raw logs, normalized timeline, analyst interpretation, evidence priority, remediation, and portfolio discussion depth.

FINAL PORTFOLIO STATEMENT - A false positive is not simply an alert that looked harmless. Here, the SIEM asserted a specific malicious network behavior that the raw source log disproved and that Wireshark, Falcon EDR, and Sysmon independently failed to corroborate. The investigation identifies the exact parser/rule defect and produces a measurable detection-engineering fix.